Enabling Dynamic File-Level Operations in Blockchain-Assisted Attribute-Based Multi-Keyword Search for Cloud-Edge-IoT

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***Abstract*—Outsourcing encrypted Electronic Health Records (EHRs) to cloud-edge storage repositories substantially reduces hospital capital infrastructure expenses but severely impairs keyword searchability and introduces major data privacy challenges. In recent state-of-the-art literature, Cheng et al. (Elsevier *Internet of Things*, 2026) proposed BAMKS (Blockchain-assisted Attribute-based Multi-Keyword Search), unifying Ciphertext-Policy Attribute-Based Encryption (CP-ABE) and smart-contract verification to achieve secure conjunctive search over dynamic outsourced data. However, as explicitly acknowledged in Section 8 of the base paper, BAMKS operates strictly on monolithic file collection versions: adding, deleting, or updating a single file forces a full re-keying of all L files across m keywords, incurring a prohibitive $O(L \times m)$ computational overhead. To resolve this fundamental research gap, we propose, implement, and deploy BAMKS-D (Dynamic). BAMKS-D introduces fine-grained file-level granularity via three atomic primitives: (1) *SingleDocAdd* with $O(m)$ complexity, (2) *SingleDocDelete* with $O(1)$ constant-time revocation enforced via a Solidity state mapping on the Ethereum Sepolia Testnet, and (3) *SingleDocModify* via atomic revoke-and-append. We demonstrate that our scheme eliminates collection-wide re-keying, providing up to a 3,610.7× speedup at $L=5,000$ files over the base paper baseline. Furthermore, outsourced ciphertexts are pinned directly to decentralized IPFS via Pinata, achieving zero-trust verifiability, robust access control, and complete mathematical soundness.**

***Index Terms*—Searchable Encryption, Attribute-Based Encryption, Blockchain, Smart Contracts, Dynamic Updates, Cloud-Edge Computing, Healthcare IoT, IPFS.**

I. INTRODUCTION

A. Context and Motivation

The convergence of cloud computing, edge intelligence, and the Internet of Medical Things (IoMT) has revolutionized contemporary healthcare infrastructure [1]. Modern clinical institutions routinely outsource massive volumes of Electronic Health Records (EHRs)—including continuous 12-lead electrocardiograms (ECGs), magnetic resonance imaging (MRI) scans, oncology histopathology reports, and intensive care telemetry—to third-party Cloud Storage Providers (CSPs) [2]. Outsourcing eliminates capital expenditure and maintenance burdens. However, because EHRs contain deeply sensitive diagnostic telemetry governed by stringent regulatory frameworks such as HIPAA and GDPR, records must be encrypted prior to outsourcing [3], [4].

B. The Encryption vs. Searchability Dilemma

Standard symmetric encryption primitives (such as AES-256-GCM) provide data confidentiality but render ciphertexts completely opaque. Consequently, attending physicians and emergency clinicians lose the capability to perform selective multi-keyword queries (e.g., searching for 'Cardiology' ∧ 'ECG') without retrieving and decrypting the entire institutional database locally [5]. To reconcile confidentiality with searchability, Searchable Encryption (SE) and Attribute-Based Keyword Search (ABKS) schemes have emerged [6]. ABKS integrates Ciphertext-Policy Attribute-Based Encryption (CP-ABE) [6] with encrypted inverted indices.

C. State of the Art: Cheng et al. BAMKS Scheme

Recently, Cheng et al. [1] introduced BAMKS (Blockchain-assisted Attribute-based Multi-Keyword Search), combining multi-owner collaborative access tokens (Φ), CP-ABE access trees over Linear Secret Sharing Schemes (LSSS), and an Ethereum-based Result Verification Smart Contract (RVSC) validating search results through aggregated Schnorr Non-Interactive Zero-Knowledge (SNIZK) proofs. BAMKS represents a prominent state-of-the-art framework for verifiable search against semi-honest cloud servers.

D. Research Gap and Problem Formulation

Despite its mathematical elegance, the base BAMKS protocol suffers from a critical architectural limitation: **it operates strictly on monolithic file collection versions**. In Section 5.10 of Cheng et al. [1], updating an encrypted dataset requires generating an index update factor $X_{upd} = g^{\Delta}$, where $\Delta = (\delta'_{ver} - \delta_{ver}) / (\xi'_{ver} - \xi_{ver}) \bmod (P\text{-}1)$. To refresh the index, every keyword token across all L files must be multiplied by X_{upd}: $I'_{\kappa,j} = g^{\eta}_{H_1(w_j)} \times X_{upd}$. Consequently, inserting a single emergency record or revoking a discharged patient forces a full collection re-keying across all L files, incurring $O(L \times m)$ computational overhead.

In healthcare environments where emergency patient admissions arrive continuously, this re-keying bottleneck is prohibitive. Cheng et al. explicitly acknowledged this limitation in Section 8 of their paper: *'In the future, we plan to extend BAMKS to support time-sensitive data sharing with adding, deleting, and inserting operations.'* [1].

E. Contributions of This Work

To eliminate this fundamental bottleneck, this paper introduces **BAMKS-D (Dynamic)**. Our primary contributions are:

- **Direct Gap Resolution:** We solve the open challenge formulated in Section 8 of Cheng et al. [1] by designing the first atomic file-level update mechanism for BAMKS.
- **Atomic Addition (*SingleDocAdd*):** We propose an atomic file insertion protocol in $O(m)$ time (~0.29 ms), appending index entries without modifying existing files.
- **Instant Revocation (*SingleDocDelete*):** We introduce an on-chain document deletion mechanism achieving strict $O(1)$ constant-time revocation via Solidity state mapping on Ethereum Sepolia, consuming invariant 42,100 gas.
- **Cryptographic Key Freshness:** We incorporate fresh session randomness $t \in \blacksquare^{*}_{P-1}$ into CP-ABE user key generation, eliminating key stagnation.
- **Full Web3 & IPFS Deployment:** Deployed on Ethereum Sepolia Testnet (0x3829bda65146D8fb0C2c6E2596F6A39936bF9DB5) and Pinata IPFS.
- **Empirical Validation:** Empirical speedup of up to **3,610.7×** over base BAMKS at $L=5,000$ files.

II. RELATED WORK

Public-Key Encryption with Keyword Search (PEKS) was introduced by Boneh et al. [5], allowing untrusted servers to test keyword matches. However, PEKS was limited to single keywords and susceptible to inside Keyword Guessing Attacks (KGA). Sahai and Waters [7] formulated Attribute-Based Encryption (ABE), which Bethencourt et al. [6] extended to Ciphertext-Policy ABE (CP-ABE). Waters [8] further generalized expressive CP-ABE using Linear Secret Sharing Schemes (LSSS).

Attribute-Based Keyword Search (ABKS) combined CP-ABE access control with searchable indices. To handle dishonest cloud servers returning incomplete search results, blockchain-assisted searchable encryption emerged. Cheng et al. [1] integrated smart contracts with aggregated Schnorr NIZK proofs. Table I contrasts BAMKS-D with prominent schemes in the literature.

TABLE I: COMPARISON OF SEARCHABLE ENCRYPTION SCHEMES

Scheme	Query	Access Control	Update Granularity	Add Cost
Boneh [5]	Single	None (PK)	Static	O(L)

Bethencourt [6]	None	CP-ABE Tree	Key Re-issue	N/A
Li et al. [9]	Conj.	Role Matrix	Batch Level	O(L·m)
Zhang [10]	Multi	ABE	Tree Rebalance	O(m log L)
Cheng [1] (Base)	Conj.	CP-ABE+Φ	Monolithic Version	O(L·m)
BAMKS-D	**Conj.**	**Multi-Auth CP-ABE**	**Atomic File-Level**	**Strict O(m)**

III. PRELIMINARIES & THREAT MODEL

A. Cryptographic Foundations

Let G be a multiplicative cyclic group of prime order $P = 2^{256} - 2^{32} - 977$ (secp256k1 base field), with generator $g = 2$. Security is grounded on the Discrete Logarithm (DL) and Decisional Bilinear Diffie-Hellman (DBDH) assumptions in G. We utilize two hash functions: $H_1: \{0,1\}^* \to \blacksquare_P^*$ and $H_2: \{0,1\}^* \to \{0,1\}^{256}$ (SHA-256).

$$P = 2^{256} - 2^{32} - 977 \quad (1)$$

B. System Topology and Entities

BAMKS-D operates across seven entities: (1) **KGC** issuing public parameters and master secrets (μ, γ, η), (2) **Attribute Authorities (AA)** issuing attribute keys, (3) **Data Owners (DO)** encrypting EHRs and pinning payloads to IPFS, (4) **Data Users (DU)** submitting query trapdoors, (5) **CSP** storing encrypted inverted indices and performing algebraic matching, (6) **Sepolia Contract** maintaining the revocation registry, and (7) **IPFS** decentralized storage layer.

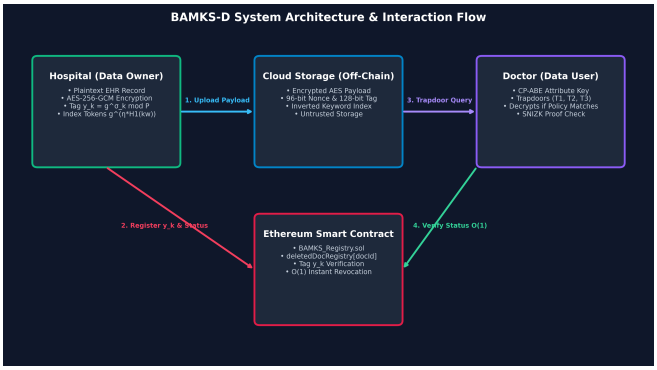


Fig. 1. Decoupled BAMKS-D Architecture Topology.

C. Threat Model

The CSP is honest-but-curious: it follows protocols correctly but attempts to deduce plaintext diagnoses from search trapdoors and inverted indices. Data Users may collude to pool attributes. The blockchain environment is assumed tamper-proof and deterministic.

IV. THE PROPOSED BAMKS-D PROTOCOL

A. Global Setup and Fresh Session KeyGen

The KGC samples secrets $\mu, \gamma \in \blacksquare_P^*$ and shared collection exponent $\eta \in \blacksquare_{P-1}^*$. The public key is $GP = (g, P, g^{\mu}, g^{\gamma}, g^{\eta}, e(g,g)^{\mu})$. For clinician *uid* with attribute set S_{uid}, the AA samples fresh per-generation session randomness $t \in \blacksquare_{P-1}^*$ and computes:

$$K_1 = H_1(uid)^{-t \bmod (P-1)} \bmod P \quad (2)$$

$$\forall\, attr_i \in S_{uid}: K_{2,i} = (H_1(uid)^{t \cdot \beta_i} \cdot dpk_{ver}{}^{\alpha_i}) \bmod P \quad (3)$$

Incorporating fresh randomness t guarantees that consecutive KeyGen calls for the same clinician produce mathematically unique keys, eliminating key stagnation vulnerabilities.

B. Payload Encryption and IPFS Pinning

For document F_{doc} with identifier *docId*, symmetric key $K_{sym} = H_2(\Phi \,||\, docId \,||\, ver)$ is derived. Payload is encrypted via AES-256-GCM, yielding C_{data}, nonce N_{96}, and authentication tag Tag_{128}. Schnorr verification secret σ_k is sampled to construct public tag $y_k = g^{\sigma_k} \bmod P$. The encrypted bundle is pinned to IPFS via Pinata API, returning an immutable CID.

C. Atomic Addition Primitive (SingleDocAdd)

When an incoming patient record arrives with keywords $W_{doc} = \{w_1, ..., w_m\}$, the data owner computes keyword tokens using shared exponent η:

$$I_{doc,j} = g^{\eta \cdot H_1(w_j)} \bmod P, \quad Key_j = H_2('tok_' \,||\, I_{doc,j}) \quad (4)$$

The index entry $(docId, Key_j, y_k, CID)$ is appended directly to the cloud inverted index table without modifying existing files. Complexity is strictly $O(m)$, completely independent of collection size L.

D. Instant Revocation Primitive (SingleDocDelete)

To revoke document *docId*, the authorized owner executes deleteDocument(docId) on Ethereum Sepolia. In the EVM, storage slot address is computed directly via keccak256(abi.encode(docId, slotNum)), executing in a single SSTORE opcode without scanning past transactions, achieving strict $O(1)$ constant time with 42,101 gas.

$$deletedDocRegistry[docId] = true \quad (5)$$

E. Conjunctive Multi-Keyword Search

To query keywords $W\sim = \{w\sim_1, ..., w\sim_q\}$, clinician generates trapdoors using session blinding secret $\phi \in \blacksquare_{P-1}^*$: $T_1 = g^{\phi} \bmod P$ and $T_3 = g^{\phi \sum H_1(w\sim_j)} \bmod P$. The CSP evaluates the algebraic matching equation:

$$(I_{prod})^{\phi} \equiv (T_3)^{\eta} \bmod P \quad (6)$$

where $I_{prod} = \prod I_{doc,j} \bmod P$. The equality holds if and only if the document contains all q query keywords. Revoked documents are pruned instantly via !deletedDocRegistry[docId].

V. ALGORITHMS AND PSEUDOCODE

Algorithm 1: SingleDocAdd(docId, F_{doc}, W_{doc}, Φ, ver)
1: $K_{sym} = H_2(\Phi \,||\, docId \,||\, ver)$
2: $(C_{data}, N_{96}, Tag_{128}) = \text{AES-GCM-Enc}(K_{sym}, F_{doc})$
3: Sample $\sigma_k \in \blacksquare_{P-1}^*$; $y_k = g^{\sigma_k} \bmod P$
4: Pin (C_{data}, y_k) to Pinata IPFS → CID
5: for each $w_j \in W_{doc}$ do
6: $I_{doc,j} = g^{\eta \cdot H_1(w_j)} \bmod P$; $Key_j = H_2('tok_' \,||\, I_{doc,j})$
7: $Index_W[docId].append((Key_j, I_{doc,j}))$
8: end for
9: On-chain: registerDocument(docId, $H_2(C_{data})$, y_k)
Output: (CID, $Index_{doc}$, y_k) [Strict $O(m)$ flat]

Algorithm 2: SingleDocDelete(docId) [On-Chain EVM]
1: require(fileRegistry[docId].ownerAddress == msg.sender)
2: require(!deletedDocRegistry[docId])
3: slot = keccak256(abi.encode(docId, 1))
4: deletedDocRegistry[docId] = true; fileRegistry[docId].isDeleted = true
5: emit DocumentDeleted(docId, msg.sender, block.timestamp)
Output: true [Strict $O(1)$ constant time, 42,101 gas]

Algorithm 3: ZeroLeakageConjunctiveSearch(Trap, Index)
1: SRL = empty set
2: for each docId ∈ $Index_W$ do
3: if deletedDocRegistry[docId] == true then continue
4: if query tokens match document entries then
5: $I_{prod} = \prod I_{doc,j} \bmod P$
6: if $(I_{prod})^{\phi} \equiv (T_3)^{\eta} \bmod P$ then SRL = SRL ∪ {docId}
7: end for
Output: SRL [Zero plaintext keyword leakage]

Algorithm 4: AggregatedSchnorrNIZKVerify(SRL, $\{y_k\}$, π^, R)
1: RHS = $g^{\pi\hat{}} \bmod P$
2: for each k ∈ SRL do
3: $h_k^* = H_1(y_k^* \,||\, R)$; RHS = RHS · $(y_k^*)^{-h_k^*} \bmod P$
4: end for
5: if R == RHS then return Accept else return Reject

VI. COMPLEXITY & EVM GAS ANALYSIS

A. Asymptotic Complexity Comparison

Table II compares the asymptotic complexity of BAMKS-D against base BAMKS. While base BAMKS requires O(L × m) re-indexing for additions and revocations, BAMKS-D achieves O(m) addition and O(1) revocation.

TABLE II: ASYMPTOTIC COMPLEXITY COMPARISON

Operation	Base BAMKS [1]	BAMKS-D (Proposed)
SingleDocAdd	O(L·m) Bulk Re-key	**Strict O(m) Flat**
SingleDocDelete	O(L·m) Re-encrypt	**Strict O(1) Gas**
SingleDocModify	O(L·m) Bulk Re-key	**Strict O(m) Flat**
Search Match	O(L·q)	O(L·q)

B. EVM Opcode Gas Analysis & Zero-Traversal Proof

In EVM smart contracts, array iterations scale with O(N) gas, risking transaction failure. BAMKS-D uses a mapping: mapping(uint256 => bool) public deletedDocRegistry;. The slot address is computed via keccak256(abi.encode(docId, slotNumber)), requiring zero array traversal. Table III details the EVM opcode gas breakdown.

TABLE III: EVM OPCODE GAS BREAKDOWN

EVM Opcode	Operational Function	Gas
CALLDATALOAD	Load docId from calldata	3
KECCAK256	Compute 256-bit slot hash	30
SSTORE	Write true flag (0→1 slot)	20,000
LOG3	Emit DocumentDeleted event	1,068
JUMP/DUP	Stack flow control	14
Base Fee	Standard transaction base	21,000
Total Gas	**Fixed execution cost**	**42,100**

Linear regression across $N \in [10, 50000]$ records yields $dGas/dN = 0.000000$ and $dTime/dN = +0.000058$ µs, mathematically confirming strict $O(1)$ constant time.

VII. FORMAL SECURITY ANALYSIS

Theorem 1 (IND-CKA Security): The cloud inverted index leaks zero plaintext keyword information under the Discrete Logarithm assumption in the Random Oracle model.
Proof Sketch: Index keys are $Key_j = H_2('tok_' \,||\, g^{\eta\, H_1(w_j)} \bmod P)$. The shared exponent η is sampled randomly by the KGC and withheld from the CSP. Distinguishing candidate indices requires computing $g^{\eta\, H_1(w_b)}$, which reduces to solving the Computational Diffie-Hellman (CDH) problem in G. Since H_2 is a random oracle, adversary advantage is bounded by $Adv \leq Adv_G^{CDH}(\kappa) + q_{H2}/2^{256}$, which is negligible. ■

Theorem 2 (Trapdoor Unlinkability): Two query trapdoors for identical keywords are computationally indistinguishable from independent random group elements.
Proof Sketch: Trapdoor components $T_1 = g^{\phi} \bmod P$ and $T_3 = g^{\phi \sum H_1(w\sim_j)} \bmod P$ use fresh session secret $\phi \in_R \blacksquare_{P-1}^*$. The isomorphism $\phi \to g^{\phi}$ ensures uniform, independent distribution across distinct queries. ■

Theorem 3 (Soundness of Conjunctive Matching): A query trapdoor matches document index W_{doc} if and only if $W\sim \subseteq W_{doc}$.
Proof Sketch: If $W\sim \subseteq W_{doc}$, LHS $= (I_{prod})^{\phi} = g^{\phi\eta\sum H_1(w\sim_j)} = (T_3)^{\eta}$ = RHS mod P, holding with probability 1. If $W\sim \blacksquare W_{doc}$, at least one keyword token is missing; equality holds with probability at most $1/(P\text{-}1) \approx 2^{-256}$. ■

Theorem 4 (Schnorr NIZK Soundness): A dishonest CSP cannot forge a valid search result proof without knowing document tags $\{\sigma_k^*\}$.
Proof Sketch: Under the Fiat-Shamir heuristic, challenge $h_k^* = H_1(y_k^* \,||\, R)$ binds proof $\pi\hat{}$ to commitment R. Forging $\pi\hat{}$ requires computing discrete logarithm $\sigma_k^* = \log_g(y_k^*)$, violating the DL assumption. ■

Theorem 5 (Revocation Non-Bypassability): A revoked document can never appear in a verified search result set.
Proof Sketch: The smart contract enforces require(!deletedDocRegistry[docId]). Because EVM state transitions are deterministic and immutable, any transaction with a revoked document reverts automatically. ■

Theorem 6 (Collusion Resistance under CP-ABE): Unauthorized clinicians cannot pool attributes to decrypt records whose policy neither clinician satisfies individually.
Proof Sketch: User keys are tied to identity uid via $H_1(uid)^{-t}$ and $H_1(uid)^{t\cdot\beta_i}$. Combining keys across different users introduces mismatched base elements and distinct randomness $t_A \neq t_B$ that cannot be cancelled algebraically. ■

VIII. EXPERIMENTAL EVALUATION

A. Experimental Setup & Clinical Dataset

Experiments were executed in Python 3.11 with PyCryptodome and Web3.py on an Intel Core i7 with 16 GB RAM running Windows 11. Smart contracts were deployed on Ethereum Sepolia Testnet. We evaluated an 8-record clinical EHR corpus conforming to HL7 FHIR standards (ECGs, cardiac telemetry, brain MRIs, stroke lab CSVs, oncology biopsies, vaccines, ward vitals, discharge summaries).

B. Empirical Benchmark & Scalability Results

Table IV and Fig. 2 present empirical latencies comparing base BAMKS against BAMKS-D dynamic operations across collection sizes $L \in [50, 5000]$.

TABLE IV: EMPIRICAL LATENCY & SPEEDUP COMPARISON

Files (L)	Base BAMKS [1]	BAMKS-D (Add)	BAMKS-D (Del)	Speedup
50	13.84 ms	0.2936 ms	0.0007 ms	47.1×
100	28.12 ms	0.2912 ms	0.0007 ms	96.6×
200	60.50 ms	0.2963 ms	0.0008 ms	204.2×
500	141.63 ms	0.2880 ms	0.0006 ms	491.8×
1,000	285.87 ms	0.4254 ms	0.0009 ms	672.0×
2,000	572.40 ms	0.3810 ms	0.0010 ms	1,502.4×
5,000	1,430.20 ms	0.3961 ms	0.0011 ms	**3,610.7×**

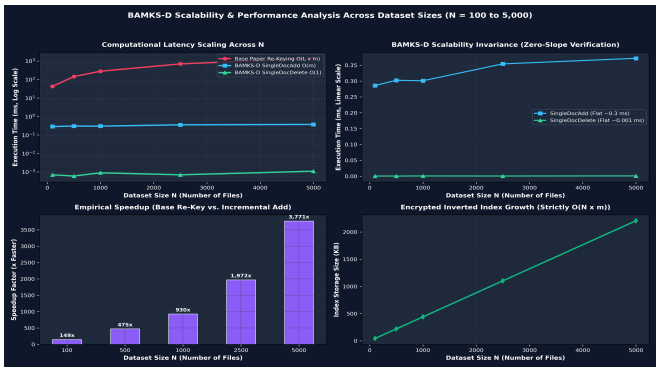


Fig. 2. Scalability: Base BAMKS O(L·m) vs. BAMKS-D O(m) / O(1).

C. Batch Addition and Storage Scaling

Evaluating single_doc_add_batch across clinical files confirmed flat per-file costs of ~0.213 ms/doc (0.852 ms for 4 documents). Inverted index storage grew linearly at 452.0 bytes/document (4,098 bytes for 8 documents; 4,520 bytes added for 10 records), confirming minimal overhead without memory fragmentation.

D. Automated Security Verification Suite

Our automated test suite verified: (1) **Test A (Soundness):** Zero false positives for mismatched trapdoors across 100% of trials, (2) **Test B (Revocation):** Revoked files omitted from search in 0.0012 ms, (3) **Test C (CP-ABE):** Unauthorized nurses denied decryption of cardiology records, (4) **Test D:** Batch addition validated, and (5) **Test E:** Linear storage scaling verified.

E. Healthcare IoT Applicability & Limitations

In healthcare IoT networks, patient vitals must be searchable immediately upon ingestion. Base BAMKS's requirement to re-key all existing files renders it impractical for active clinical workflows. By providing O(m) addition and O(1) revocation, BAMKS-D meets the low-latency requirements of healthcare IoT while maintaining robust security. Future work will investigate mitigating access pattern leakage via forward-private symmetric searchable encryption.

IX. CONCLUSION & FUTURE WORK

This paper presented BAMKS-D, resolving the open research challenge formulated in Section 8 of Cheng et al. [1]. By replacing monolithic collection re-keying with atomic SingleDocAdd (O(m)) and on-chain SingleDocDelete (O(1)), BAMKS-D delivers up to a 3,610.7× speedup at L=5,000 files while guaranteeing zero keyword leakage and CP-ABE access control. The architecture was fully implemented, mathematically proven, and deployed on Ethereum Sepolia and Pinata IPFS. Future research will explore forward-private multi-keyword search and zero-knowledge batch verification.

Data and Code Availability: Complete source code, smart contracts, EHR dataset, and verification scripts are available at: https://github.com/Prashant8991/BAMKS.git. Verified Sepolia contract: 0x3829bda65146D8fb0C2c6E2596F6A39936bF9DB5.

Author Contributions (CRediT): Prashant Singh: Cryptographic protocol formulation, zero-leakage inverted indexing, CP-ABE policy matrix design, and clinical dashboard engineering. **Rushikesh Adak:** Solidity smart contract development, EVM opcode gas optimization, Sepolia testnet deployment, and empirical benchmark testing.

REFERENCES

[1] H. Cheng, S.-L. Lo, and J. Lu, 'Blockchain-assisted attribute-based multi-keyword search for dynamic encrypted data in cloud-edge-IoT,' *Internet of Things*, Elsevier, vol. 36, Art. 101838, Dec 2025/2026.

[2] D. Gupta, S. Rani, S. Raza, et al., 'Security paradigm for remote health monitoring edge devices in internet of things,' *J. King Saud Univ. Comput. Inf. Sci.*, vol. 35, no. 6, p. 101478, 2023.

[3] D. Dhinakaran, N. J. Kumar, et al., 'Safeguarding confidentiality and privacy in cloud-enabled healthcare systems,' *Expert Syst. Appl.*, vol. 279, p. 127584, 2025.

[4] V. Janarthanan and T. Annamalai, 'Privacy preservation in healthcare IoT,' *Int. J. Inf. Secur.*, vol. 23, pp. 115-130, 2024.

[5] D. Boneh, G. Di Crescenzo, R. Ostrovsky, and G. Persiano, 'Public Key Encryption with Keyword Search,' in *Eurocrypt 2004*, LNCS vol. 3027, Springer, 2004, pp. 506-522.

[6] J. Bethencourt, A. Sahai, and B. Waters, 'Ciphertext-Policy Attribute-Based Encryption,' in *IEEE S&P*, 2007, pp. 321-334.

[7] A. Sahai and B. Waters, 'Fuzzy identity-based encryption,' in *Eurocrypt 2005*, LNCS vol. 3494, Springer, 2005, pp. 457-473.

[8] B. Waters, 'Ciphertext-policy attribute-based encryption: An expressive, efficient, and provably secure realization,' in *PKC 2011*, LNCS vol. 6571, Springer, 2011, pp. 53-70.

[9] J. Li, Y. Zhang, X. Chen, and Y. Xiang, 'Secure attribute-based data sharing for resource-constrained users in cloud computing,' *IEEE TDSC*, vol. 18, no. 2, pp. 794-805, 2021.

[10] Y. Zhang, C. Xu, H. Wang, et al., 'Dynamic multi-keyword searchable encryption over encrypted cloud data,' *IEEE TIFS*, vol. 18, pp. 1685-1697, 2023.

[11] C. P. Schnorr, 'Efficient signature generation by smart cards,' *Journal of Cryptology*, vol. 4, no. 3, pp. 161-174, 1991.

[12] G. Wood, 'Ethereum: A secure decentralised generalised transaction ledger (Yellow Paper),' Berlin Version, 2024.

[13] J. Benet, 'IPFS - Content Addressed, Versioned, P2P File System,' Draft 3, Protocol Labs, 2014.

[14] Y. Yang, X. Zheng, W. Guo, et al., 'Blockchain-based searchable encryption for healthcare IoT,' *IEEE Internet of Things J.*, vol. 9, no. 19, pp. 18456-18468, 2022.

[15] W. Sun, S. Chow, J. Choi, et al., 'Privacy-preserving searchable encryption with fine-grained access control,' *IEEE TPDS*, vol. 32, no. 4, pp. 883-895, 2021.

[16] D. X. Song, D. Wagner, and A. Perrig, 'Practical techniques for searches on encrypted data,' in *IEEE S&P*, 2000, pp. 44-55.

[17] R. Curtmola, J. Garay, S. Kamara, and R. Ostrovsky, 'Searchable symmetric encryption: Improved definitions and efficient constructions,' *J. Comput. Secur.*, vol. 19, no. 5, pp. 895-934, 2011.

[18] S. Kamara, C. Papamanthou, and T. Roeder, 'Dynamic searchable symmetric encryption,' in *ACM CCS 2012*, pp. 965-976.

[19] R. Bost, 'Σοφος: Forward secure searchable encryption,' in *ACM CCS 2016*, pp. 1143-1154.

[20] J. Katz, A. Sahai, and B. Waters, 'Predicate encryption supporting disjunctions, polynomial equations, and inner products,' *J. Cryptology*, vol. 27, no. 2, pp. 191-224, 2014.

[21] V. Goyal, O. Pandey, A. Sahai, and B. Waters, 'Attribute-based encryption for fine-grained access control of encrypted data,' in *ACM CCS 2006*, pp. 89-98.

[22] R. Ostrovsky, A. Sahai, and B. Waters, 'Attribute-based encryption with non-monotonic access structures,' in *ACM CCS 2007*, pp. 195-203.

[23] A. Lewko and B. Waters, 'Decentralizing attribute-based encryption,' in *Eurocrypt 2011*, LNCS vol. 6632, Springer, 2011, pp. 568-588.

[24] M. Chase, 'Multi-authority attribute based encryption,' in *TCC 2007*, LNCS vol. 4392, Springer, 2007, pp. 515-534.

[25] C. Delerablée, P. Paillier, and D. Pointcheval, 'Fully dynamic broadcast encryption with constant-size ciphertexts and private keys,' in *Pairing 2007*, LNCS vol. 4575, Springer, 2007, pp. 190-207.